EE3110 - Probability Foundations for Electrical Engineers
Tutorial - 6

1. Let X_1 and X_2 be independent Poisson random variables with parameters $\lambda_1 > 0$ and $\lambda_2 > 0$, respectively. Define $X = X_1 + X_2$. Find the p.m.f. of X. Find the expected value and variance of X.

2. Suppose that $X_1, X_2, \ldots$ are i.i.d. (independent and identically-distributed) random variables, each of which have the moment generating function $\psi(t)$. Let $Y = X_1 + \cdots + X_N$, where the number of terms N in this sum is a random variable having the Poisson distribution with mean λ. Assume that N and $X_1, X_2, \ldots$ are independent, and $Y = 0$ if $N = 0$. Determine the m.g.f. of Y.

3. Let N be a geometric random variable with parameter p and X_i, $i = 1, 2, \ldots$, be *i.i.d* Poisson random variables with mean λ. Find the p.m.f of $Y = \sum_{i=1}^{N} X_i$, i.e. Y is a random variable obtained by taking a sum of geometric number of independent Poisson random variables. You can leave the answer as a summation.

4. Heights (in inches) of women in a population are normally distributed with mean 65 and standard deviation 1. Heights (in inches) of men in the same population are normally distributed with mean 68 and standard deviation 3. One woman and one man are selected independently at random. Let W denote the woman's height and M the man's height. Find the probability that the woman is taller than the man, i.e. find $\Pr(W > M)$.

   (Use the standard normal random variable Z. The Z-score you will need is $(\Phi(0.94868) \approx 0.8289)$

5. Suppose $X_1, \ldots, X_n$ are i.i.d. Bernoulli(p) and let $S = \sum_{i=1}^{n} X_i$. For $k = 1, \ldots, n$, compute $\mathbb{P}(X_1 = 1 \mid S = k)$.

6. Let $X \sim \text{Bin}(n, p)$ with probability mass function

$$f(x) = \mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

   Find the mode(s) of X, i.e. the value of x for which $f(x)$ is maximum.

7. Let X be the number of distinct birthdays in a group of 110 people (i.e., the number of days in a year such that at least one person in the group has that birthday). Under the usual assumptions (no February 29, all the other 365 days of the year are equally likely, and the day when one person is born is independent of the days when the other people are born), find the mean and variance of X.

8. A total of 20 bags of Haribo gummy bears are randomly distributed to the 20 students in a certain Stat 110 section. Each bag is obtained by a random student, and the outcomes of who gets which bag are independent. Find the average number of bags of gummy bears that the first three students get in total, and find the average number of students who get at least one bag.

9. (a) Do there exist two discrete random variables X and Y such that X and Y have the same distribution, i.e, same PMF and CDF, but the event $X = Y$ never occurs? If yes, give an example. Assume same sample space.

   (b) Let's consider a simplified gacha mechanic, where you spend *tickets* to obtain two types of rewards: a Standard reward and a Premium reward. In one *pull*, you spend one *ticket* to receive a reward, which is either Standard or Premium with equal probability 1/2. The reward from a *pull* is independent of other *pulls*. You decide to keep *pulling* till you obtain at least one Standard and one Premium reward and then stop. What is the expected number of *pulls*? Assume that in each *pull* a reward is definitely obtained and that you have infinite *tickets*.

1. Let X_1 and X_2 be independent Poisson random variables with parameters $\lambda_1 > 0$ and $\lambda_2 > 0$, respectively. Define $X = X_1 + X_2$. Find the p.m.f. of X. Find the expected value and variance of X.

$X = X_1 + X_2$

$P(X_1 = k) = \frac{e^{-\lambda_1} \lambda_1^k}{k!}$ $\quad P(X_2 = l) = \frac{e^{-\lambda_2} \lambda_2^l}{l!}$

$P(X = n) = P(X_1 + X_2 = n)$

$= \sum_{k=0}^{n} P(X_1 = k) \cdot P(X_2 = n-k)$

$= \sum_{k=0}^{n} \frac{e^{-\lambda_1} \lambda_1^k}{k!} \cdot \frac{e^{-\lambda_2} \lambda_2^{n-k}}{(n-k)!}$

$= \frac{e^{-(\lambda_1+\lambda_2)}}{x!} \sum_{k=0}^{x} {}_nC_k (\lambda_1)^k (\lambda_2)^{x-k}$

$= \frac{e^{-(\lambda_1+\lambda_2)}}{x!} (\lambda_1 + \lambda_2)^x = \text{poisson}(\lambda_1 + \lambda_2)$

$E[X] = \sum_{x=0}^{\infty} x \cdot \frac{e^{-(\lambda_1+\lambda_2)}}{x!} (\lambda_1+\lambda_2)^x = (\lambda_1+\lambda_2) \sum_{x=1}^{\infty} \frac{e^{-(\lambda_1+\lambda_2)} (\lambda_1+\lambda_2)^{x-1}}{(x-1)!}$

$= \lambda_1 + \lambda_2$

$E[X^2] = \sum_{x=0}^{\infty} x^2 \frac{e^{-(\lambda_1+\lambda_2)}}{x!} (\lambda_1+\lambda_2)^x = (\lambda_1+\lambda_2)(1+\lambda_1+\lambda_2)$

$\sigma = (\lambda_1+\lambda_2)^2 + (\lambda_1+\lambda_2) - (\lambda_1+\lambda_2)^2 = (\lambda_1+\lambda_2)$

3. Let N be a geometric random variable with parameter p and X_i, $i = 1, 2, \ldots$, be $i.i.d$ Poisson random variables with mean λ. Find the p.m.f of $Y = \sum_{i=1}^{N} X_i$, i.e. Y is a random variable obtained by taking a sum of geometric number of independent Poisson random variables. You can leave the answer as a summation.

$P(X_i = n) = e^{-\lambda} \lambda^n / n!$ $\quad P(N = n) = p(1-p)^{n-1}$

$X_1 + X_2 + \ldots + X_n \sim e^{-(n\lambda)} (n\lambda)^n / n!$

$P(Y = k) = \sum_{n=1}^{\infty} P(Y = k / N = n) \cdot P(N = n)$

$$= \sum_{n=1}^{\infty} e^{-(n\lambda)} (n\lambda)^x / x! \cdot p(1-p)^{n-1}$$

5. Suppose $X_1, \ldots, X_n$ are i.i.d. Bernoulli(p) and let $S = \sum_{i=1}^n X_i$. For $k = 1, \ldots, n$, compute $\mathbb{P}(X_1 = 1 \mid S = k)$.

$$P(X_1=1 \mid S=K) = \frac{P(X_1=1 \cap S=K)}{P(S=K)}$$

$$P(S=K) = {}^nC_K p^K (1-p)^{n-K}$$

$$P\left(\sum_{i=2}^{n} X_i = K-1\right) = {}^{n-1}C_{K-1} \cdot p^{K-1}(1-p)^{n-K} \qquad P(X_1=1)=p$$

$$\Rightarrow \frac{p \cdot {}^{n-1}C_{K-1} p^{K-1} (1-p)^{n-K}}{{}^nC_K \, p^K (1-p)^{n-K}} = \frac{K}{n}$$

6. Let $X \sim \text{Bin}(n, p)$ with probability mass function

$$f(x) = \mathbb{P}(X = x) = \binom{n}{x} p^x (1-p)^{n-x}, \qquad x = 0, 1, \ldots, n.$$

Find the mode(s) of X, i.e. the value of x for which $f(x)$ is maximum.

$$f(x) = {}^nC_x \, p^x (1-p)^{n-x}$$

$$f(x-1) = {}^nC_{x-1} \, p^{x-1} (1-p)^{n-x+1}$$

$$\frac{f(x)}{f(x-1)} = \frac{n!}{(n-x)!\,x!} \, p^x (1-p)^{n-x} \cdot \frac{(x-1)!\,(n-x+1)!}{n! \cdot p^{x-1} \cdot (1-p)^{n-x+1}}$$

$$= \frac{p(n-x+1)}{x(1-p)} \geq 1$$

$$\Rightarrow pn - px + p \geq x - px \Rightarrow x \leq p(n+1)$$

$(n+1)p \notin \mathbb{Z} \Rightarrow$ mode is $\lfloor (n+1)p \rfloor = x$

$(n+1)p \in \mathbb{Z} \Rightarrow$ mode is $x = (n+1)p$ & $(n+1)p - 1$

2. Suppose that $X_1, X_2, \ldots$ are i.i.d. (independent and identically-distributed) random variables, each of which have the moment generating function $\psi(t)$. Let $Y = X_1 + \cdots + X_N$, where the number of terms N in this sum is a random variable having the Poisson distribution with mean λ. Assume that N and $X_1, X_2, \ldots$ are independent, and $Y = 0$ if $N = 0$. Determine the m.g.f. of Y.

$P(N=n) = e^{-\lambda}\lambda^n/n!$

$$E[e^{tY}] = E[e^{t(X_1+\cdots+X_n)}] = \sum_{n=0}^{\infty} [\psi(t)]^n \cdot \frac{e^{-\lambda}\lambda^n}{n!}$$

$$= e^{-\lambda}\left[\sum_{n=0}^{\infty} \frac{[\lambda\psi(t)]^n}{n!}\right] = e^{-\lambda}\cdot e^{\lambda\psi(t)} = e^{\lambda(\psi(t)-1)}$$

4. Heights (in inches) of women in a population are normally distributed with mean 65 and standard deviation 1. Heights (in inches) of men in the same population are normally distributed with mean 68 and standard deviation 3. One woman and one man are selected independently at random. Let W denote the woman's height and M the man's height. Find the probability that the woman is taller than the man, i.e. find $\Pr(W > M)$.

(Use the standard normal random variable Z. The Z-score you will need is ($\Phi(0.94868) \approx 0.8289$)

$M \sim N(68, 9)$ $\quad W \sim N(65, 1)$

$A = W - M > 0$ $\quad A \to$ normal variable as D, W are indep...

$E[A] = E[W-M] = -3$

$Var(A) = Var(W-M) = 9+1 = 10$

$\Rightarrow A \sim N(-3, 10)$ $\quad Z = \frac{x-\mu}{\sigma}$

$$P(A>0) = P\left(Z > \frac{0-(-3)}{\sqrt{10}}\right) = P\left(Z > \frac{3}{\sqrt{10}}\right) = P(Z > 0.948)$$

Given $P(Z \le 0.948) = 0.8289 \Rightarrow$ Req $= 0.1711$

7. Let X be the number of distinct birthdays in a group of 110 people (i.e., the number of days in a year such that at least one person in the group has that birthday). Under the usual assumptions (no February 29, all the other 365 days of the year are equally likely, and the day when one person is born is independent of the days when the other people are born), find the mean and variance of X.

$N = 365$; $n = 110$ $\quad x \to$ no of distinct bdays

$I_j = 1 \Rightarrow$ atleast 1 guy has b-day on day j

$$X = \sum_{j=1}^{N} I_j$$

$$E[X] = E\left[\sum_{j=1}^{N} I_j\right] = \sum_{j=1}^{N} E[I_j] = \sum_{j=1}^{N} P(I_j = 1)$$

$$= N - \sum_{j=1}^{N} P(I_j = 0)$$

$$= 365 - 365\left(\frac{364}{365}\right)^{110}$$

$$\text{Var}(X) = \text{Var}\left(\sum_{j=1}^{N} I_j\right) = \sum_{j=1}^{N} \text{Var}(I_j) + \sum_{j \neq k} \text{Cov}(I_j, I_k)$$

$$365\left(1 - \left(\frac{364}{365}\right)^{110}\right)\left(\frac{364}{365}\right)^{110}$$

$$\text{Cov}(I_j, I_k) = E[I_j I_k] - E[I_j] \cdot E[I_k]$$

$$P(I_j = 1 \cap I_k = 1) \qquad \left(1 - \left(\frac{364}{365}\right)^{110}\right)^2$$

$$= 1 - P(I_j = 0 \cup I_k = 0)$$

$$1 - P(I_j = 0) - P(I_k = 0) + P(I_j = 0 \cap I_k = 0)$$

$$1 - 2\times\left(\frac{364}{365}\right)^{110} + \left(\frac{363}{365}\right)^{110}$$

$$= \left[\left(\frac{363}{365}\right)^{110} - \left(\frac{364}{365}\right)^{220}\right] 365 \cdot 364$$

$$\text{Var}(X) = 365\left(1 - \left(\frac{364}{365}\right)^{110}\right)\left(\frac{364}{365}\right)^{110} + 365 \cdot 364\left[\left(\frac{363}{365}\right)^{110} - \left(\frac{364}{365}\right)^{110}\right]$$

8. A total of 20 bags of Haribo gummy bears are randomly distributed to the 20 students in a certain Stat 110 section. Each bag is obtained by a random student, and the outcomes of who gets which bag are independent. Find the average number of bags of gummy bears that the first three students get in total, and find the average number of students who get at least one bag.

$x_i \rightarrow$ no of bags student i gets

$$E[x_1 + x_2 + x_3] = E[x_1] + E(x_2) + E(x_3)$$

Given a bag, prob of student i getting it is $1/20$

$X_i \sim Bin(20, 1/20) \Rightarrow E[X_i] = 20 \cdot 1/20 = 1$

$\Rightarrow E[X_1 + X_2 + X_3] = 3$

$I_j \rightarrow$ student j get a bag

$$E\left[\sum_{j=1}^{20} I_j\right] = \sum_{j=1}^{20} E[I_j] = \sum_{j=1}^{20} P(I_j = 1) = 20 - \sum_{j=1}^{20} P(I_j = 0)$$

$$= 20 - 20\left(\frac{19}{20}\right)^{20}$$

$$= 20\left(1 + \left(\frac{19}{20}\right)^{20}\right) \approx 12.83$$

9. (a) Do there exist two discrete random variables X and Y such that X and Y have the same distribution, i.e, same PMF and CDF, but the event $X = Y$ never occurs? If yes, give an example. Assume same sample space.

   (b) Let's consider a simplified gacha mechanic, where you spend *tickets* to obtain two types of rewards: a Standard reward and a Premium reward. In one *pull*, you spend one *ticket* to receive a reward, which is either Standard or Premium with equal probability 1/2. The reward from a *pull* is independent of other *pulls*. You decide to keep *pulling* till you obtain at least one Standard and one Premium reward and then stop. What is the expected number of *pulls*? Assume that in each *pull* a reward is definitely obtained and that you have infinite *tickets*.

(a) $\Omega = \{1, 2, 3, 4\}$ $\quad \omega \in \Omega$

ω	1	2	3	4
$X(\omega)$	0	1	0	1
$Y(\omega)$	1	0	1	0

$P(X=0) = 1/2$

$P(Y=0) = 1/2$

$P(X=Y) = 0$

(b) $N \rightarrow$ no of pulls $\quad S \rightarrow$ one Std atleast $\quad P$

$$E[N] = E[N/X_1 = S] \cdot P[X_1 = S] + E[N/X_1 = P] \cdot P[X_1 = P]$$

$P[X_1 = S] = 1/2$, $P[X_1 = P] = 1/2$

$\swarrow$ 1+ additional how many for P (geometric)

$= 1 + 1/(1/2) = 3$

$E[N] = 3/2 + 3/2 = 3$

10. **2D transformation of a bivariate Gaussian RV:** Let the joint PDF of two Gaussian random variables X, Y be given as

$$f_{XY}(x,y) = \frac{1}{2\pi(1-\rho^2)} \exp\left[-\left(\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right)\right].$$

Compute the joint PDF $f_{VW}(v,w)$, where

$$V = \frac{1}{2}(X^2 + Y^2),$$

$$W = \frac{1}{2}(X^2 - Y^2).$$

11. Let $N \sim \text{Poisson}(\lambda)$. Conditional on $N = n$, define

$$X \mid (N = n) \sim \text{Binomial}(n, p).$$

Find the marginal distribution of X and show that

$$X \sim \text{Poisson}(p\lambda).$$

12. Let (X, Y) have a bivariate normal distribution. Which of the following statements must be true? 1. Any nonzero linear combination of X and Y has a normal distribution. 2. $E(Y \mid X = x)$ is a linear function of x. 3. $\text{Var}(Y \mid X = x) \leq \text{Var}(Y)$.

13. Let random variables Y_1, Y_2, Y_{12} be independent, where

$$Y_i \sim \text{Poisson}(\lambda_i), \quad i \in \{1, 2, 12\}.$$

Define

$$X_1 = Y_1 + Y_{12}, \qquad X_2 = Y_2 + Y_{12}.$$

(a) Find the covariance $\text{Cov}(X_1, X_2)$.

(b) Derive the joint probability mass function (pmf) of (X_1, X_2).

(c) Comment on the independence of X_1 and X_2 when $\lambda_{12} = 0$.

10. **2D transformation of a bivariate Gaussian RV:** Let the joint PDF of two Gaussian random variables X, Y be given as

$$f_{XY}(x,y) = \frac{1}{2\pi(1-\rho^2)} \exp\left[-\left(\frac{x^2 - 2\rho xy + y^2}{2(1-\rho^2)}\right)\right].$$

Compute the joint PDF $f_{VW}(v,w)$, where

$$V = \frac{1}{2}(X^2 + Y^2),$$

$$W = \frac{1}{2}(X^2 - Y^2).$$

$x^2 + y^2 = 2v$

$xy = \pm\sqrt{v^2 - w^2}$

$x = \pm\sqrt{v+w} \quad y = \pm\sqrt{v-w} \qquad v \geq |w|$

$$J = \begin{pmatrix} \frac{dv}{dx} & \frac{dv}{dy} \\ \frac{dw}{dx} & \frac{dw}{dy} \end{pmatrix} = \begin{pmatrix} x & y \\ x & -y \end{pmatrix} \Rightarrow |J| = -2xy$$

$$|J^{-1}| = \frac{1}{2|xy|} = \frac{1}{2\sqrt{v^2 - w^2}}$$

$$f_{VW}(v,w) = \sum_{i=1}^{4} f_{XY}(x_i, y_i)|J^{-1}|$$

$$f_{VW}(v,w) = \frac{1}{\pi(1-\rho^2)} \left\{ \frac{e^{-\left[\frac{v + \rho\sqrt{v^2-w^2}}{(1-\rho^2)}\right]} + e^{-\left[\frac{v - \rho\sqrt{v^2-w^2}}{(1-\rho^2)}\right]}}{2\sqrt{v^2 - w^2}} \right\}$$

$$= \frac{e^{-v/(1-\rho^2)} \cosh\left[\frac{\rho\sqrt{v^2-w^2}}{1-\rho^2}\right]}{\pi(1-\rho^2)\sqrt{v^2-w^2}}$$

11. Let $N \sim \text{Poisson}(\lambda)$. Conditional on $N = n$, define

$$X \mid (N = n) \sim \text{Binomial}(n, p).$$

Find the marginal distribution of X and show that

$$X \sim \text{Poisson}(p\lambda).$$

$$P(X=k) = \sum_{n=0}^{\infty} P(X=k \mid N=n) \cdot P(N=n)$$

$$= \sum_{n=0}^{\infty} {}^{n}C_k \, p^k (1-p)^{n-k} \cdot \frac{e^{-\lambda}\lambda^n}{n!}$$

$$= \sum_{n=0}^{\infty} \frac{n!}{(n-k)!\,k!} p^k (1-p)^{n-k} \cdot \frac{e^{-\lambda}\lambda^n}{n!}$$

$$= \left(\frac{p}{1-p}\right)^k \frac{e^{-\lambda}}{k!} \sum_{n=0}^{\infty} \frac{(1-p)^n \lambda^n}{(n-k)!}$$

$$= \lambda^k p^k \frac{e^{-\lambda}}{k!} \sum_{n=0}^{\infty} \frac{(1-p)^{n-k}\lambda^{n-k}}{(n-k)!} = \frac{(\lambda p)^k e^{-\lambda}}{k!} \cdot e^{\lambda(1-p)}$$

$$= e^{-\lambda p} (\lambda p)^k / k! = \text{Poisson}(\lambda p)$$

12. Let (X, Y) have a bivariate normal distribution. Which of the following statements must be true? 1. Any nonzero linear combination of X and Y has a normal distribution. 2. $E(Y \mid X = x)$ is a linear function of x. 3. $\text{Var}(Y \mid X = x) \leq \text{Var}(Y)$.

1. True

$\mu_Z = a\mu_X + b\mu_Y \qquad \sigma_Z^2 = a^2\sigma_X^2 + b^2\sigma_Y^2$

2. $Y/X=x \sim \text{Normal}\left(\mu_{Y/X=x}, \sigma^2_{Y/X=x}\right)$

$E[Y/X=x] = \mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X)$ $\rho \to$ correlation

True

3. $\text{Var}(Y|X=x) = \sigma_Y^2(1-\rho^2)$ True

13. Let random variables Y_1, Y_2, Y_{12} be independent, where

$$Y_i \sim \text{Poisson}(\lambda_i), \quad i \in \{1, 2, 12\}.$$

Define

$$X_1 = Y_1 + Y_{12}, \qquad X_2 = Y_2 + Y_{12}.$$

(a) Find the covariance $\text{Cov}(X_1, X_2)$.

(b) Derive the joint probability mass function (pmf) of (X_1, X_2).

(c) Comment on the independence of X_1 and X_2 when $\lambda_{12} = 0$.

$\text{cov}(Y_i, Y_j) = 0 \quad i \neq j$

(a) $\text{cov}(X_1, X_2) = \text{cov}(Y_1+Y_{12}, Y_2+Y_{12})$

$= \text{cov}(Y_1, Y_2) + \text{cov}(Y_1, Y_{12}) + \text{cov}(Y_2, Y_{12}) + \text{cov}(Y_{12}, Y_{12})$

$= 0 + 0 + 0 + \text{var}(Y_{12}) = \lambda_{12}$

(b) $P(X_1=x_1, X_2=x_2) = \sum_{k=0}^{\min(x_1,x_2)} P(X_1=x_1, X_2=x_2 \mid Y_{12}=k)$

$\cdot P(Y_{12}=k)$

$\downarrow$

$P(Y_1=x_1-k, Y_2=x_2-k \mid Y_{12}=k)$

$= P(Y_1=x_1-k)\ P(Y_2=x_2-k)$

$$\sum_{k=0}^{\min(x_1,x_2)} \frac{e^{-\lambda_1}\lambda_1^{x_1-k}}{(x_1-k)!} \frac{e^{-\lambda_2}\lambda_2^{x_2-k}}{(x_2-k)!} \frac{e^{-\lambda_{12}}\lambda_{12}^{k}}{k!}$$

$$= e^{-(\lambda_1+\lambda_2+\lambda_{12})} \cdot \lambda_1^{x_1}\lambda_2^{x_2} \sum_{k=0}^{\min(x_1,x_2)} \frac{\lambda_{12}^{k}}{(x_1-k)!\,(x_2-k)!\,k!\,\lambda_1^{k}\lambda_2^{k}}$$

$x_1, x_2 \in \{0,1,2\ldots\}$

(c) $\lambda_{12}=0 \Rightarrow f_{X_1X_2}(x_1,x_2) = \dfrac{e^{-(\lambda_1+\lambda_2)}\lambda_1^{x_1}\lambda_2^{x_2}}{x_1!\,x_2!}$ = independent